

# Preparative Native PAGE Purification of Membrane Bound Active Nitrate Reductase from *Geobacter Metallireducens* using a Modified Model 491 Prep Cell

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## Abstract

*Geobacter metallireducens* is a relatively newly discovered dissimilatory iron reducing bacterium that has a potential use in bioremediation. Besides iron, it can also grow by dissimilatory reduction of nitrate to ammonia through a novel reductase system (Naik *et al.*, 1993). This report describes the use of native gel electrophoresis with a modified Bio-Rad Model 491 Prep Cell system for the purification of the active nitrate reductase involved in the nitrate reduction pathway of this bacterium. With this system we have been able to completely purify the nitrate reductase and a c-type cytochrome associated with the nitrate reduction process.

The Model 491 Prep Cell has been modified by omitting the inner cooling core unit, pumps, and the collection system. This modification minimizes the loss of protein by diffusion caused by an otherwise very long conventional run, allows for a larger sampling surface, and maintains the resolution of the body of the gel. The electrophoresis is stopped before the dye front runs off the gel. The gel is sliced and the different fractions electro-eluted and analyzed for activity. The system is used as the disk electrophoresis of Ornstein and Davis (1964), but with a much broader resolving surface.

With this technique, we have been able to consistently isolate and purify very hydrophobic, membrane bound, active enzymes that were difficult to isolate by conventional chromatography and FPLC (e.g., gel filtration and ion exchange chromatography columns). We believe this technique can be very useful in the future for the purification of other such proteins.

## Introduction

The strict anaerobe, *Geobacter metallireducens*, has been characterized as a dissimilatory iron reducing bacterium which couples the oxidation of acetate to the reduction of ferric iron (Lovley and Phillips, 1988; Lovley *et al.*, 1993). It also oxidizes a variety of short chain fatty acids and aromatic compounds, and can use nitrate, Mn(IV), or U(VI) as terminal electron acceptors (Lovley *et al.*, 1993). This type of organism is very

important in the oxidation of organics and the reduction of inorganics and metals in anaerobic soils, aquifers, and estuarine sediments (Lovley, 1991). Nevertheless, its biochemistry is poorly understood. Our goal was to purify the nitrate reductase and cofactors involved in nitrate reduction. The nitrate reductase, like other tightly bound membrane proteins, was difficult to solubilize and purify by conventional procedures. We tried gravity chromatography and FPLC, with anion and cation exchange columns, hydrophobic interaction chromatography, and gel filtration, but none of them gave significant successful purification. In addition, as a consequence of the large size of this enzyme, its poor solubility, and its propensity to aggregate, we were not able to adequately purify sufficient quantities for analysis by conventional continuous elution preparative gel electrophoresis. The use of native gel electrophoresis with a modification of the standard Bio-Rad Model 491 Prep Cell methodology made it possible to isolate, purify, and obtain the enzyme in active form.

## Methods

### Culture Conditions

*Geobacter metallireducens*, strain GS-15 (ATCC 53774), was grown anaerobically using the Hungate technique (Hungate, 1969) in freshwater acetate medium, supplemented with 20 mM NaNO<sub>3</sub> (final electron acceptor) as described in Lovley and Phillips (1988).

All cultures were grown to mid-logarithmic phase and harvested. Membrane and soluble cytoplasmic fractions were prepared by sonication and differential centrifugation, using a method modified from Stolz *et al.* (1990) and Ketchum *et al.* (1991). All fractions were stored at -20 °C.

### Sample Preparation

After harvesting and fraction separation, 1.5 ml of the membrane fraction, with a total protein content of about 15 mg, was incubated with 0.5 ml 2% CHAPS (in phosphate buffer, pH 7.2) and 0.5 ml electrophoresis sample buffer (1.5 M Tris, pH 6.8, 0.5% (v-v), 0.4% glycerol (v-v), 0.5% CHAPS) for 7 min at 70 °C. At that point, they were loaded into the native gel electrophoresis apparatus (see next page).

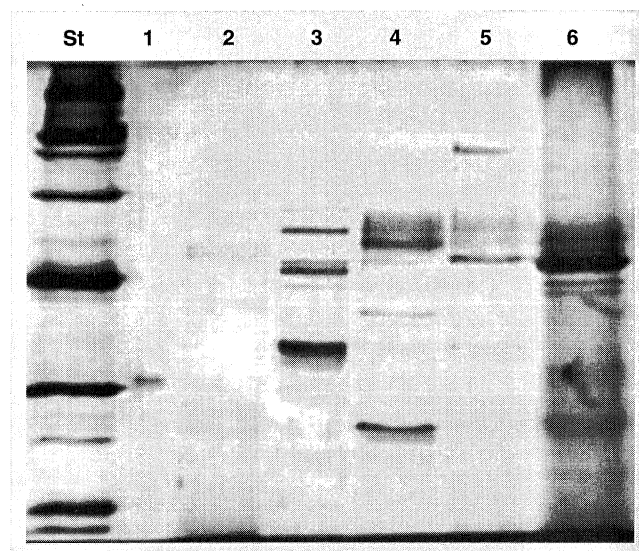
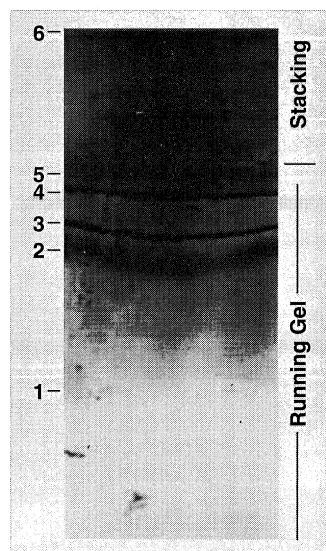
## Preparative Gel Electrophoresis

We used the Bio-Rad Model 491 Prep Cell system, in which no cooling core unit or pumps were used and no fractions were eluted from the gel. The set-up employs only the upper and lower buffer chambers, gel assembly tube, and elution chamber without a dialysis membrane and with the cooling buffer port closed with a stopper. The frits were still in place to support the gel, but there was no need for dialysis membrane, since no proteins were eluted. We had exactly the same successful results running the system without the elution chamber, using only the elution frit attached to the fluted gasket of the gel assembly tube with a thin band of Parafilm® laboratory film on the periphery. The upper buffer chamber contains buffer only in the inner space; the outer chamber can remain empty. The cooling buffer inlet valve is closed, as is the cooling buffer line.

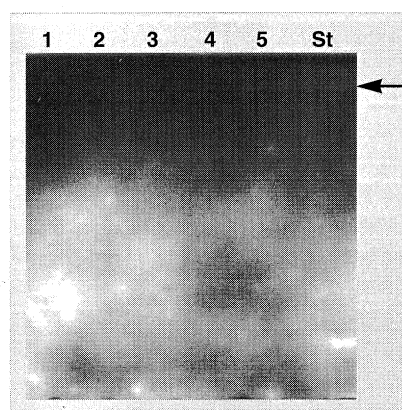
A 7.0 % polyacrylamide resolving gel (5 cm) and a 4% polyacrylamide stacking gel (1.5 cm) were cast in the small (28 mm ID) gel tube assembly without the cooling core and left to polymerize in the cold (10 °C) for about 1 hr. Native gel electrophoresis electrode buffer (37 mM tris and 0.28 M glycine, with 0.1% CHAPS in the upper running buffer) was used throughout the electrophoresis run. The Model 491 Prep Cell was run at 150 V for 3.5 hr in a cold chamber (at about 10 °C), until the dye front was just about to run off the gel. The electrophoresis was then stopped, and the polyacrylamide cylinder extruded from the gel assembly tube. This was done by carefully rinsing the walls with water, using a syringe and needle, until the whole gel had separated from the glass. The gel then came out smoothly from the tube.

In order to identify and obtain our proteins of interest, their distribution in the gel was previously determined by running mini slab gel native electrophoresis under the same conditions as used with the Model 491 Prep Cell. We detected our proteins and ascertained their migration pattern by enzyme activity assays and silver staining of the mini gels. An additional way to find the protein distribution in the Prep Cell was by cutting and staining a thin longitudinal slice all along the gel (see Figure 1).

**Fig. 1. Pattern of protein migration in a longitudinal slice of a solid 28 mm Prep Cell cylinder gel.** The electrophoresis was stopped at a point where the proteins bands were still sharp. The gel was stained with Coomassie® blue. 1 through 6 are the bands observed.



**Fig. 2. Protein content of each band from the Prep Cell, after electro-elution.** St is molecular weight standards, lanes 1 through 6 correspond to the bands from Figure 1 sliced and electro-eluted. SDS-PAGE, 11.5%, stained with silver nitrate.



**Fig. 3. Activity stain (methyl viologen assay) of samples extracted from the Prep Cell.** Lanes 1 through 5 correspond to the Bands sliced and electro-eluted from those in Figure 1, St is a control for the enzyme activity (arrow).

Subsequently, slices from the Model 491 Prep Cell gel (about 3 mm thick) were then cut at the expected distance from the start of the resolving gel and the migration front. In the slicing process, enough area of the gel was covered in order to insure maximum recovery of our proteins and that no protein was left behind. After that, the slices were treated independently. In order to estimate their protein content, the slices were ground (to facilitate the elution of proteins), and electro-eluted overnight using the same native electrophoresis buffer used in the Model 491 Prep Cell. The electro-eluted samples were then analyzed in a mini SDS-PAGE gel to determine their degrees of purity or number of proteins, and tested for enzyme activity with the methyl viologen assay adapted for the non-denaturing electrophoresis gel (Blumle and Zumft, 1991) (Figures 2 and 3).

## Results and Discussion

A modified Model 491 Prep Cell proved successful in purifying membrane proteins that were otherwise difficult to purify by traditional chromatography methods and elution electrophoresis.

Neither the cooling core unit nor pumps were used in the system. Instead, the electrophoresis was performed in a cold chamber (10 °C) to avoid overheating problems. Attempts to use the large (37 mm ID) gel tube assembly gave problems with overheating, even in the cold chamber.

In our system, no fractions are eluted continuously from the gel. The high molecular weight of the native complex (in excess of 400 kD), and the very hydrophobic nature of our proteins of interest, strongly retarded their electrophoretic migration even in low percentage acrylamide gels. Furthermore, they also became highly diffused soon after separation. The longer the run, the greater the diffusion; hence the necessity of a run long enough for separation, but not so long as to allow diffusion of the proteins. Thus we stopped the electrophoresis at an early phase, which in our case was just before the dye front ran off the gel.

The resulting system is comparable with large format slab gel electrophoresis, and also to the disk gel of Ornstein and Davis, (1964). The advantages over the former are the convenience of the gel handling and the higher sample loading surface. Heat-generated band spreading was minimized with the use of a cold chamber for the polymerization and electrophoresis.

In conclusion, the advantages of our system are the ease of assembly and operation, and the important increase in both resolving surface and sample amount loaded. The result is the purification, in sufficient quantities, of even membrane proteins of low abundance.

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